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# Does risk premium help uncover the uncovered interest parity failure?

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ABSTRACT

There is a general consensus emerging that the uncovered interest parity (UIP) does not hold or ex-post exchange rate change predicts the interest rate differential in the wrong direction. This paper provides a pioneer study to offer a risk premium based solution to the popular UIP failure using a dataset of 44 advanced and emerging currencies. We report that in the absence of risk premium, UIP failure is more prominent in emerging countries relative to advanced countries since only 34% of the total beta coefficients range between 0.5 and 1.5. Next, we include the risk premium in the main UIP equation using a component generalized autoregressive conditional heteroskedastic-in-mean (CGARCH-M) model and show that the results conform more to the UIP theory since 73% of the beta coefficients range between 0.5 and 1.5. Such a finding validates our argument that risk premium is the main factor responsible for UIP violation and including it in the main equation helps uncover the UIP puzzle, especially in the case of emerging countries. Overall, in the presence of risk premium, in most countries, ex-post exchange rate change bear a positive relationship with the interest rate differential as UIP predicts. This is our key contribution to the literature.



## 1. Introduction

The uncovered interest rate parity (UIP) theory postulates that a high interest rate currency would depreciate against those with lower interest rates such that the higher returns in high interest rate currencies are compensated by the equivalent loss associated with the currency depreciation. In other words, the ex-post exchange rate change should equal the interest rate differential between two countries (or a regression of ex-post exchange rate change on interest rate differential should generate a beta coefficient close to one). On the contrary, empirical research has reported a large and negative beta coefficient implying that high interest rate currencies generally have a tendency to appreciate, particularly over a short-term horizon (Aysun and Lee, 2014; Cho and Chun, 2019). This is the UIP failure, a condition when the exchange rate change is negatively related to interest rate differential (or forward rate is not an unbiased predictor of the future spot rate), popularly known as *forward premium puzzle* (Fama, 1984). This puzzle has now become a stylized fact and has been reported in many

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studies (see, for example, Fama, 1984; Kaminsky and Peruga, 1990; Backus et al., 1995; Li et al., 2012; Kumar and Trück, 2014; Kumar, 2018, Bussiere et al., 2018; Coulibaly and Kempf, 2019, etc.).

A number of studies have attempted to provide a rationale for the UIP failure through different arguments[1]; they however, fall in two main categories. The first category argues that the systematic component of the UIP failure or forward rate prediction error is due to the errors in the expectations of market participants (Campbell et al., 2007; Bacchetta and van Wincoop, 2010).[2] The second argument is associated with the assumption of rational expectations, and the component of the *puzzle* is attributed to a risk premium (Frankel and Poonawala, 2010; Kumar and Trück, 2014). The risk premium based explanation for the failure of UIP is also found in Fama (1984), Inci and Lu (2007), Li et al. (2012), Loring and Lucey (2013), and Ismailov and Rossi (2018). They argue that failure of the UIP happens when the expected excess returns on currencies, often referred to as risk premiums, are possibly not zero. This situation arises when risk averse investors present in the market demand a risk premium causing the forward rate to deviate from the future spot rate. For the empirical evidence to fit the theory, Fama (1984) argues that these time varying risk premiums should be negatively correlated with the expected depreciation of the home currency, and exhibit a relatively higher variance. Since the asset pricing models have failed to generate risk premium with such characteristics, the puzzle has become even more prominent (Ismailov and Rossi, 2018).

Many studies examining the UIP condition in the context of emerging and advanced countries have produced mixed results with a lack of consensus on how to explain the failure. Engel (1996) presents an excellent survey of the literature on this puzzle. Lothian (2016) argues that the UIP condition holds well for a sample of 17 countries across two centuries. Bansal and Dahlquist (2000), while studying this relation for a sample of 28 countries, propose that UIP failure is a common phenomenon in developed countries, not in developing ones. Similar results are also reported in Frankel and Poonawala (2010). Flood and Rose (2002) and Aysun and Lee (2014) show that UIP deviations are significant both in developed and developing countries. However, Francis et al. (2002), and Ahmad et al. (2012) show that UIP deviations in developing nations are more pronounced than advanced ones. Adewuyi and Ogebe (2019) test the UIP condition in the African countries and show that the risk premiums do not revert towards UIP equilibrium, therefore, UIP does not hold in these countries. Ferreira and León-Ledesma (2007) argue that risk premium seems to be the appropriate specification particularly for emerging countries. Therefore, we focus on examining the potential role of risk premiums in uncovering the UIP failure in advanced and emerging economies, especially over long-run.

In this paper, we compare the UIP failure and the explanatory power of risk premium across a group of 22 advanced and 22 emerging countries for a better comparison of UIP since the previous studies have mixed results for these two sets of countries. The emerging countries are characterized by lower per capita income, higher inflation and uncertainty, and therefore, higher interest rates. It gives us sufficient reasons to believe that these economic differences would directly influence the relation between exchange rate change and interest rate differentials. Frankel and Poonawala (2010) show that emerging market currencies are riskier compared to other developed currencies, therefore, risk premium would be higher and more variable than other advanced currencies. They further argue that if the bias is higher in emerging markets (higher likelihood of UIP failure), it should be interpreted as an evidence of higher risk premium. Bansal and Dahlquist (2000) show that the negative exchange rate change and interest rate differential relation has direct implication for the expected excess returns (risk premiums) obtained from investing in foreign assets. Based on the different risk-return tradeoff in the currency deposits in emerging and advanced countries, we argue that the likelihood of UIP failure in emerging countries should be higher compared to the advanced countries. Francis et al. (2002) also show that a large component of the UIP failure in developing markets is attributed to the risk premium. Therefore, comparing the UIP and risk premium results across emerging and advanced currencies is crucial to better comprehend the dynamics of exchange rates across these nations.

Based on the arguments above, we believe that risk premium is an important component for UIP failure. We use the risk premium in the UIP equation and contribute to the literature by providing a solution to the UIP failure or *forward premium puzzle*. We argue that the Asian financial crisis and the 2008 global financial crisis (GFC) could play an important role as far as the impact of risk premium on UIP is concerned. Following these crises, a rapid change in risk across the world has been observed (Li et al., 2012). Therefore, we include the risk premium in the main UIP equation by employing a component generalized autoregressive conditional heteroskedastic in mean (CGARCH-M) model to account for changes in risks across the world due to their inherent risk characteristics. Doing so would enable us separate the risk in the UIP equation into permanent and transitory components which is crucial to understanding whether the risk inherent in UIP is governed by major macroeconomic forces or short-term market dynamics. Therefore, this model appears to be superior to other GARCH models in that it can differentiate between the permanent and temporary components of exchange rates volatility.

Another unique feature of our work is that we provide an extensive examination of the UIP condition over a span of almost five decades. Although, at short horizons, the large empirical evidence argue in the favor of UIP failure, the recent studies find the evidence that UIP holds better at longer horizons, as ours (Chinn and Meredith, 2004; Lothian and Wu, 2011; Moore and Roche, 2012). Bekaert et al. (2007) and Mehl and Cappiello (2009) argue that UIP failure might happen

[1] For instance, Engel et al. (2019) show that including inflation in the UIP equation renders the interest rate differential insignificant while the inflation variable is found to be highly significant. Similarly, Coulibaly and Kempf (2019) show that those countries, which implement the inflation targeting, witness a violation of the UIP condition.

[2] There have been other explanations for UIP failure provided in the literature. For instance, Gourinchas and Tornell (2004) and Burnside et al. (2010) provide behavioral explanations; Carlson et al. (2008) put forward institutional factors; dos Santos et al. (2016) argue for transaction cost arising from hedging activities. Cho and Chun (2019) show that that the degree of the persistence of the forward premium plays an important role in explaining the puzzle.

at short horizons, not at longer horizon because market frictions in the short run do not let exchange rates respond immediately to the interest rate changes. Similarly, Cheung et al. (2019) show that in short-run, the random walk model of the exchange rate outperforms the interest rate differentials, however, at longer horizons, the interest differential performs better than a random walk, consequently, the UIP theory holds well at longer horizons.

One of the studies which come closer to our work is Li et al. (2012), who use the CGARCH-M model for the first time to test the UIP theory. Our study is different from theirs at least on three counts. *First*, our dataset is broader (44 currencies) compared to theirs (10 currencies). *Second*, we cover a much longer period in contrast to Li et al. (2012), whose analysis covers the data only up to 2009. Their main argument focuses on the rapid change in risk across the world, especially after the 2008 financial crisis; however, their data do not cover much of the post-crisis data to validate the argument. They further recommend that future research should analyze a longer span of data across a wide range of currencies. *Third* and the most important, though, Li et al. (2012) find significant risk premiums for many developed and developing currencies, however, unlike the results we obtain, their findings are not in line with the UIP since their beta estimates were not unambiguously positive even after applying the GARCH-M model. We add more to the extant literature by addressing the UIP anomaly. In particular, we argue that risk premium is responsible for UIP failure especially for emerging countries and that including it in the UIP framework helps uncover the UIP failure over longer horizons. The key explanation that we are able to address the UIP failure compared to Li et al. (2012) might be due to the fact that during the recent times, the capital has become more mobile among emerging and advanced economies as a result of the development of international financial markets.

Our results indicate that unlike advanced currencies, most of the beta coefficients for emerging countries are negative suggesting an inverse relation between the expected exchange rate change and interest rate differential, a condition of UIP failure. We show that only 34% of the beta coefficients for both advanced and emerging countries lie between 0.5 and 1.5, implying that UIP failure is a common phenomenon, especially among emerging countries, when the risk premium is not accounted for. However, using a CGARCH-M model in the main UIP equation, we find that risk premium is more significant for emerging currencies. Noticeably, in the presence of risk premium, the beta coefficients move closer to one and lie between 0.5 and 1.5 in 73% of the cases for both the advanced and emerging currencies. This evidence suggests that in the presence of risk premium, a positive interest rate differential leads to depreciation of the home currency, conforming to the UIP theory. Further, the estimates of the CGARCH-M model show that the time-varying risk premium is largely driven by long-term macroeconomic fundamentals, not by short-term market sentiments. Overall, we conclude that inclusion of risk premium in the UIP equation helps uncover the UIP failure and argue that risk premium should be considered in all theoretical and empirical models of exchange rates.

Our findings provide implications for exchange rate models that try to explain the forward premium puzzle. In Fama (1984), the negative relation between the expected depreciation and the interest rate differential suggests the negative relation between risk premium and expected depreciation. Further, Hodrick and Srivastava (1986), and Backus et al. (1998) argue that the negative correlation between the risk premium and expected depreciation can be rationalized by simple asset pricing models. However, several studies have failed to address the higher variance of risk premium. For instance, the monetary models of Lucas (1982), and models that incorporate trading frictions in goods markets (Hollifield and Uppal, 1997) fail to account for the forward premium puzzle. We argue for the additional imperative dimensions in the expected depreciation rates and interest rate differential relation—the risk premium. In particular, we attempt to understand the UIP failure in the presence of risk premium in emerging and advanced economies.

The rest of the paper has been organized as follows: Next section demonstrates the research methodology used and the underlying theory. Section 3 shows the data and preliminary results, and Section 4 explains the main empirical result. Finally, Section 5 concludes.

## 2. Research methodology – The UIP theory and the risk premium

The UIP condition states that

$$(E_t s_{t+k} - s_t) = i_{t,k} - i^*_{t,k} \tag{1}$$

where $E_t s_{t+k}$ is the natural log of the expected spot rate of the home country's currency against the foreign country's currency at $t + k$. $i_{t,k}$ ($i^*_{t,k}$) is the interest rate for the home country's (foreign country) bonds at time $t$ maturing in time $k$. According to the theory of covered interest rate parity,[3]

$$(f^k_t - s_t) = i_{t,k} - i^*_{t,k} \tag{2}$$

where $f^k_t$ represents the natural log of the forward rate at time $t$ for maturity in time $k$. Combining Eqs. (1) and (2) leads to forward rate unbiasedness hypothesis (FRUH) as

$$E_t s_{t+k} - s_t = f^k_t - s_t \tag{3}$$

[3] Akram et al. (2008) and Pippenger (2011) state that the theory of covered interest rate parity holds true for all time in the currency markets.

For testing the unbiasedness hypothesis in the currency futures market, Eq. (3) will lead us to testing the typical Fama (1984) UIP equation as:

$$s_{t+k} - s_t = \alpha + \beta(f_t^k - s_t) + \varepsilon_{t+k} \tag{4}$$

The term on the left side of Eq. (4) is the change in the log of spot exchange rate ($\Delta s_{t+k}$) and ($f_t^k - s_t$) is the forward premium or the basis. Hence, we test the null hypothesis of FRUH as $\alpha = 0$ and $\beta = 1$.

However, it is a common phenomenon in the literature that the beta coefficient in Eq. (4) is never found to be one, rather negative in number of cases. This condition is called forward premium anomaly and one of the reasons for it suggested in the literature is the time-varying risk premium. Many studies examining UIP are based on the postulation that investors are risk neutral and form rational expectations (Li et al., 2012; Kumar and Trück, 2014). Since the investors can be safely assumed to be risk averse, studies not including the risk premium are more prone to reporting a negative $\beta$ coefficient. According to Fama (1984), $f_t^k$ can be broken down into two components – expected spot exchange rate and risk premium as:

$$f_t^k = E_t s_{t+k} + \pi_{t+k} \tag{5}$$

where $\pi$ is a measure of risk premium. Subtracting $s_t$ from both sides, Eq. (5) becomes:

$$f_t^k - s_t = (E_t s_{t+k} - s_t) + \pi_{t+k} \tag{6}$$

Assuming the covered interest parity (CIP) holds ($f_t^k - s_t = i_{t,k} - i_{t,k}^*$), Eq. (6) becomes as in Li et al. (2012):

$$i_{t,k} - i_{t,k}^* = (E_t s_{t+k} - s_t) + \pi_{t+k} \tag{7}$$

Rearranging Eq. (7) leads to:

$$(E_t s_{t+k} - s_t) = i_{t,k} - i_{t,k}^* - \pi_{t+k} \tag{8}$$

Eq. (8) suggests that expected spot rate change would not be equal to the interest rate differential; rather it would be equal to the interest rate differential minus a risk premium. Since holding assets in emerging nations are relatively riskier in comparison to advanced ones, investors in emerging markets would require more compensation in terms of risk premium coupled with higher interest rates in emerging markets to attract foreign inflows. If the expectations are assumed to be rational, Eq. (8) changes as:

$$s_{t+k} - s_t = i_{t,k} - i_{t,k}^* - \pi_{t+k} + \varepsilon_{t+k} \tag{9}$$

Combining Eqs. (4) and (9), the final UIP equation appears as:

$$s_{t+k} - s_t = \alpha + \beta\left(i_{t,k} - i_{t,k}^*\right) + \theta h_{t+k} + \varepsilon_{t+k} \tag{10}$$

where $h_{t+k}$ is the error term's conditional volatility, representing the dynamic risk premium. According to Eq. (10), risk premium has two components, $\alpha$, the constant component, and $\theta$, the time-varying component. The finding of both $\alpha$ and $\theta$ being zero would confirm the absence of risk premium. If $\alpha \neq 0$ and $\theta = 0$, it implies a constant risk premium. However, if $\alpha = 0$ and $\theta \neq 0$, it is understood that there is only time-varying risk premium.

In their survey analysis, Froot and Frankel (1989) decompose the UIP deviations into two components – the expectation errors and the risk premium, where expectation errors play a more significant role in UIP failure. However, Froot and Thaler (1990) and Cavaglia et al. (1993) argue that risk premium plays more significant role relative to expectation error. Similarly, Chinn and Meredith (2004) show that especially in the short run, risk premium leads to the failure of UIP. Inci and Lu (2007) argue that the unbiasedness condition holds for the shorter maturity contracts, while it is rejected for longer maturity contracts, implying the importance of risk premium in the long-run. Similarly, Kumar and Trück (2014) show that for longer maturity contracts, risk premiums are more significant in the Indian context.

Poghosyan et al. (2008), using a GARCH-M model, substantiate a positive risk premium in Armenia. Similarly, Melander (2009) show that UIP deviations decline after the risk premium is incorporated into the GARCH-M model. Li et al. (2012) decompose the risk premium into its permanent and transitory risk components using a CGARCH-M model. They show that risk premium is a significant element for many currencies, especially in the emerging countries. Kiani (2013) shows the significant presence of risk premium in ten different currencies using a GARCH model. On one hand, Frankel and Poonawala (2010) show that UIP failure is more significant for advanced currencies relative to developing currencies. Loring and Lucey (2013) on the other hand, find the contradictory results in that the UIP failure is more significant for emerging currencies. They also show significant risk premiums for emerging currencies. However, Aysun and Lee (2014), using a GARCH-M model, show that UIP deviations are equally noteworthy in developed and emerging markets and a bigger part of them are attributable to risk premiums. dos Santos et al. (2016) apply a CGARCH-M model and show that UIP does not hold in their data and that a significant risk premium is present which is persistent in the long-term.

Recent studies have widely used CGARCH models to separate volatility into permanent and transitory components since many of the investment decisions are dependent on the nature of risk (Li et al., 2012). Owing to these reasons, many studies have argued that CGARCH models perform superior to normal GARCH models (see, for instance, Guo and Whitelaw, 2006; Li et al., 2012, etc.). The GARCH-M models were introduced by Engle et al. (1987) and further developed by Bollerslev et al. (1992) to effectively model the risk-return relationships. We follow Berk and Knot (2001) and Li et al. (2012) and employ CGARCH-M model using Eq. (11) as:

$$s_{t+k} - s_t = \alpha + \beta\left(i_{t,k} - i^*_{t,k}\right) + \theta h_{t+k} + \varepsilon_{t+k}$$

$$h^2_{t+k} = \gamma_0 + \gamma_1 \varepsilon^2_t + \gamma_2 h^2_t \tag{11}$$

where $h_{t+k}$ represent the time-varying risk premium. The main reason behind using CGARCH-M model is that it can decompose the volatility in permanent and transitory components, since volatility varies with time. The permanent or the long-term component of volatility is assumed to be driven by the macroeconomic fundamentals, while the transitory component reflects the short term market movement such as trading and other microstructure issues. Except Li et al. (2012) and dos Santos et al. (2016), no other study has used the CGARCH model to test the UIP theory. Similar to Li et al. (2012), we include an asymmetric effect in our model to understand the volatility differences during currency appreciations and depreciations as below:

$$s_{t+k} - s_t = \alpha + \beta\left(i_{t,k} - i^*_{t,k}\right) + \theta h_{t+k} + \varepsilon_{t+k}$$

$$q_{t+k} = \gamma_1 + \gamma_2(q_t - \gamma_1) + \gamma_3\left(\varepsilon^2_t - h^2_t\right) \tag{12}$$

$$h^2_{t+k} = q_{t+k} + \gamma_4(\varepsilon^2_t - q_t) + \gamma_5 D_t(\varepsilon^2_t - q_t) + \gamma_6\left(h^2_t - q_t\right)$$

where $D_t$ acts as a dummy to account for the asymmetric effects of exchange rates. For an unexpected currency appreciation, $D_t$ equals *one*, for $\varepsilon_t < 0$, otherwise, $D_t$ equals *zero*. $q_{t+k}$, the conditional variance's permanent constituent, signifies the effect of macroeconomic forces. It reaches to long-run volatility, $\gamma_1$ having a scale of $\gamma_2$ and shows the permanent characteristics of the variance. The $(h^2_t - q_t)$ signifies short-run or transitory component governed by short term market movement such as trading and other microstructure issues and is more volatile in nature. For the model to be stable, $\gamma_2$ should be more than the coefficient $(\gamma_4 + \gamma_6)$ of the transitory volatility, implying the short-run volatility converges quicker relative to the long-run volatility. The closer $\gamma_2$ is to one, the slower $q_{t+k}$ advances towards $\gamma_1$, while closer its value is to zero, the quicker $q_{t+k}$ moves towards $\gamma_1$. Therefore, $\gamma_2$ signifies the long-term persistence. $\gamma_3$, the forecast error coefficient, implies the effect of shock on the permanent or long-term component of volatility. We also incorporate a dummy variable in the model to account for the asymmetric impact of historical shocks on the short-run volatility. If $\gamma_5 < 0$, the effect of negative shocks (i.e., unexpected depreciation) on the transitory volatility is more than that of positive shocks (i.e., unexpected appreciation). To put it differently, the short-run volatility increases as a result of unexpected depreciation.

## 3. Data and preliminary results

The monthly data for exchange rates for a set of 44 currencies have been collected from *Datastream* and the interest rates have been taken from the Bank of International Settlements (BIS). Many of the developed countries had been liberalized since the 1970s, followed by some in 1980s, while emerging markets' liberalization peaked around the early and mid 1990s. We have classified the currencies in advanced and emerging currencies, 22 each, based on International Monetary Fund (IMF) country classification. The data period, as explained in Table 1, starts from a minimum of 18 years (in case of Qatar) to a maximum of 47 years for many advanced currencies, till it ends in December 2017. It is evident in the table that the data for many emerging countries was available only from early 1990s, therefore, the data for emerging markets is included as and when it becomes available. While the same data for advanced countries are available from as early as 1970s. Hence the reason behind the choice of data depends upon its availability. The considered one-month interest rates for each country are annualized. For some countries, the interest rates were not available (e.g., India, Argentina, Venezuela and Philippines), therefore, we have resorted to the use of bank rates as a substitute, similar to Bansal and Dahlquist (2000). The exchange rates are specified in terms of units of domestic currency per unit of US dollar and the value is on the first day of each month.

Table 1 also presents the key summary statistics of the log changes in exchange rates. The table shows that dollar has depreciated against half of the advanced currencies; however, it has strongly appreciated against all the emerging countries (except Brunei Darussalam). It is also evident that emerging currencies are more volatile relative to advanced economies. Table 1 further report the unit root statistics for exchange rate changes using the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests both of which show that the considered time-series is stationary for all currencies, for the considered time period.

**Table 1**
Key descriptive statistics.

| Advanced Countries | | | | | | Emerging Countries | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | SD | ADF | PP | Data from | | Mean | SD | ADF | PP | Data from |
| Australia | −0.91 | 10.85 | −48.91*** | −108.3*** | Jan-71 | Russia | 17.9 | 19.09 | −38.82*** | −56.63*** | Jul-93 |
| UK | −0.97 | 9.231 | −47.40*** | −104.9*** | Jan-71 | Colombia | 6.38 | 9.651 | −34.29*** | −72.59*** | Aug-92 |
| Taiwan | −0.61 | 4.031 | −36.27*** | −87.76*** | Feb-84 | Turkey | 23.8 | 19.02 | −35.96*** | −73.65*** | Sep-93 |
| Canada | 0.63 | 6.150 | −48.09*** | −107.8*** | Jan-71 | Brazil | 25.7 | 16.25 | −26.85*** | −73.03*** | Jan-92 |
| Switzerland | −3.14 | 11.47 | −48.27*** | −107.3*** | Jan-71 | South Africa | 6.51 | 13.90 | −49.50*** | −108.7*** | Jan-71 |
| France | 0.18 | 9.877 | −46.69*** | −106.0*** | Jan-71 | India | 4.80 | 7.400 | −47.25*** | −111.9*** | Jan-73 |
| Czech Republic | −0.64 | 11.82 | −33.99*** | −76.86*** | Jun-93 | Argentina | 9.33 | 13.03 | −33.17*** | −70.26*** | Jan-92 |
| Denmark | −0.20 | 10.12 | −48.02*** | −111.0*** | Jan-71 | Hungary | 4.98 | 13.19 | −35.38*** | −76.17*** | Jun-93 |
| Hong Kong | 0.98 | 3.518 | −44.04*** | −112.4*** | Apr-74 | Chile | 2.36 | 8.730 | −33.29*** | −71.10*** | Apr-81 |
| Sweden | 1.06 | 10.49 | −49.14*** | −108.5*** | Jan-71 | Peru | 4.12 | 5.395 | −32.52*** | −80.62*** | Aug-92 |
| Singapore | −1.10 | 5.323 | −41.62*** | −100.9*** | Jan-81 | Brunei Darussalam | −0.58 | 6.414 | −36.13*** | −80.79*** | Jun-93 |
| German | −1.52 | 10.05 | −47.40*** | −106.7*** | Jan-71 | Qatar | 0.00 | 0.346 | −43.97*** | −145.3*** | Jan-99 |
| Japan | −2.30 | 10.05 | −47.87*** | −106.8*** | Jan-71 | Saudi Arabia | 0.00 | 0.468 | −44.70*** | −128.4*** | Dec-88 |
| Greece | 4.95 | 11.09 | −41.45*** | −97.09*** | Apr-81 | Indonesia | 7.80 | 22.68 | −35.97*** | −81.37*** | Nov-91 |
| Norway | 0.41 | 10.36 | −48.64*** | −110.0*** | Jan-71 | Mexico | 15.4 | 24.93 | −48.35*** | −114.8*** | Oct-82 |
| Italy | 2.26 | 9.833 | −47.67*** | −107.9*** | Jan-71 | Poland | 3.56 | 12.87 | −35.63*** | −75.66*** | Jun-93 |
| Spain | 1.98 | 10.58 | −46.57*** | −107.7*** | Jan-73 | Malaysia | 0.69 | 6.841 | −48.03*** | −104.8*** | Jan-71 |
| Portugal | 0.76 | 10.49 | −37.42*** | −86.57*** | Dec-88 | Morocco | 2.97 | 13.62 | −36.20*** | −80.74*** | Apr-93 |
| Belgium | −0.65 | 10.07 | −46.56*** | −106.9*** | Jan-71 | Venezuela | 0.71 | 10.46 | −45.72*** | −108.8*** | Aug-92 |
| Netherlands | −1.26 | 10.01 | −46.67*** | −106.6*** | Jan-71 | Thailand | 3.03 | 9.552 | −18.94*** | −41.76*** | Jan-73 |
| South Korea | 1.55 | 11.69 | −54.51*** | −89.22*** | Apr-81 | Philippine | 8.07 | 5.223 | −30.61*** | −73.34*** | Dec-93 |
| Austria | −1.57 | 10.92 | −47.78*** | −113.8*** | Jan-71 | Bulgaria | 14.8 | 8.906 | −31.95*** | −98.06*** | Jan-91 |

**Note:** The table presents the basic characteristics of monthly exchange rate changes for a set of 22 advanced and 22 emerging currencies. Means and standard deviations have been annualized by multiplying with a factor of $12 * 100$ and $\sqrt{12} * 100$. SD denotes standard deviation. ADF stands for Augmented Dickey-Fuller test while PP shows Philips-Perron test of stationarity.
*** Shows significance at 1% level. The data period for all currencies ends in December 2019.

## 4. Main results

Table 2 reports the OLS results of Eq. (4) separately for advanced and emerging economies. Panel A shows that for advanced countries, the range of the beta coefficients is from −1.715 to 2.173 with an average beta of 0.544. Out of 22 currencies, the beta coefficient is significantly different from one only for six currencies (Taiwan, Canada, Denmark, Hong Kong, Singapore and Norway) at least at 5% level of significance. On the contrary, the results in Panel B show that the beta coefficient is negative and significantly different from zero for almost all emerging currencies (except, Russia, Poland, Malaysia and Thailand) at least at 5% level of significance. The beta coefficients range from a minimum of −3.629 to a maximum of 3.384 with an average of −0.493.

The beta coefficients from emerging countries are negative implying that a rise in interest rate differential will cause the expected exchange rate to decrease. However, a positive beta coefficient for advanced economies indicate that UIP prevails, that is, high interest rate country's currency should depreciate against those with lower interest rates. Overall, the results in Table 2 indicate that UIP holds for 16 out of 22 advanced currencies and only for four out of 22 emerging countries. Fig. 1 shows the frequency distribution of the beta coefficients and reveal that only 34% of the beta coefficients (15 out of 44) for both advanced and emerging countries lie between 0.5 and 1.5 with an average beta of 0.85.

The results in Table 2 reveal that likelihood of UIP failure is more common in emerging countries relative to advanced countries. These findings contradict the traditional results that UIP failure is a common phenomenon for advanced and developed countries. However, these findings are in line with Alexius (2001), Chinn and Meredith (2004), and Lothian (2016) who shows that UIP puzzle disappears over longer horizons. Our results also compliment the work of Loring and Lucey (2013) who show that UIP failure is less significant for advanced countries. Mehl and Cappiello (2009) also show that UIP holds well for mature and developed countries, especially at long horizons; however, there is less support for UIP in case of emerging currencies. They further argue that in case of emerging currencies, political risk and risk premiums help explain the UIP failure.

One of the possible explanations for our results could be related to the risk premium theory. The finding of $\beta$ being significantly less than zero for emerging countries imply that a rise in forward premium reduces the spot currency change, in addition to an increase in risk premium (Li et al., 2012). This in turn implies that investors would require higher risk premium as a compensation for investing in risky currencies coupled with an expectation of currency appreciation, rather than depreciation. For holding risky currencies, investors are rewarded on two fronts – currency appreciation gains and higher interest rate gains. Therefore, by virtue of risk premium, the negative relation between exchange rate change and interest rate differential is more common in emerging countries, resulting in more incidents of UIP failure.

To account for the role of risk premium in addressing the UIP failure, we next use the CGARCH-M model. Tables 3.1 and 3.2 present the results for UIP condition using a CGARCH-M model as shown in Eq. (12) assuming the generalized error

**Table 2**
OLS tests for UIP.

| Panel A: Advanced Countries | | | | | | Panel B: Emerging Countries | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\alpha$ | $\beta$ | SE | $t$, $\beta$ = 0 | $t$, $\beta$ = 1 | | $\alpha$ | $\beta$ | SE | $t$, $\beta$ = 0 | $t$, $\beta$ = 1 |
| Australia | −0.319 | 0.876 | 0.628 | 1.396 | −0.197 | Russia | 1.521 | 0.892 | 0.588 | 1.517 | −0.184 |
| UK | −0.421 | 0.793 | 0.447 | 1.773 | −0.463 | Colombia | 0.586** | −3.629** | 1.825 | −1.988 | −2.536 |
| Taiwan | −0.623** | 0.314*** | 0.193 | 1.630 | −3.553 | Turkey | 0.438 | −2.796*** | 1.208 | −2.314 | −3.141 |
| Canada | −0.743** | −1.715** | 1.115 | −1.538 | −2.434 | Brazil | 0.861 | −2.152*** | 0.977 | −2.203 | −3.227 |
| Switzerland | −0.827** | 0.607 | 0.387 | 1.568 | −1.014 | South Africa | 2.085*** | −1.626*** | 0.545 | −2.982 | −4.816 |
| France | −0.062 | 1.259 | 0.864 | 1.458 | 0.300 | India | −0.582 | 1.723*** | 0.211 | 8.157 | 3.422 |
| Czech Republic | 1.287 | 0.618 | 0.476 | 1.298 | −0.802 | Argentina | 0.839** | −1.421*** | 0.394 | −3.602 | −6.138 |
| Denmark | −0.901 | −0.542** | 0.739 | −0.733 | −2.085 | Hungary | −1.244* | −2.582*** | 0.913 | −2.828 | −3.923 |
| Hong Kong | −1.289* | −1.186*** | 0.508 | −2.332 | −4.299 | Chile | −0.768 | 3.384*** | 0.899 | 3.763 | 2.651 |
| Sweden | 0.856 | 2.173 | 0.833 | 2.609 | 1.408 | Peru | −0.954** | −0.809*** | 0.423 | −1.912 | −4.275 |
| Singapore | −1.180** | 0.436** | 0.228 | 1.912 | −2.471 | Brunei Darussalam | −0.004 | −2.057*** | 0.960 | −2.143 | −3.186 |
| Germany | −0.528 | 0.964 | 0.668 | 1.442 | −0.054 | Qatar | 0.333 | −0.885*** | 0.483 | −1.831 | −3.902 |
| Japan | −1.519** | 0.814 | 0.712 | 1.142 | −0.261 | Saudi Arabia | 0.804* | −0.846** | 0.764 | −1.108 | −2.418 |
| Greece | 0.736 | 1.986 | 0.738 | 2.689 | 1.335 | Indonesia | 0.312 | 0.208** | 0.330 | 0.628 | −2.400 |
| Norway | −0.565 | −0.441*** | 0.429 | −1.028 | −3.359 | Mexico | −0.627** | 3.148*** | 0.428 | 7.355 | 5.019 |
| Italy | 2.582 | 1.644 | 0.489 | 3.360 | 1.316 | Poland | 0.826 | 1.317 | 0.312 | 4.221 | 1.017 |
| Spain | 0.224 | 1.110 | 0.411 | 2.701 | 0.267 | Malaysia | 0.657 | 0.763 | 0.681 | 1.120 | −0.349 |
| Portugal | −0.708 | 0.873 | 0.563 | 1.550 | −0.226 | Morocco | −1.565** | −0.966*** | 0.722 | −1.338 | −2.722 |
| Belgium | −1.343* | −0.482 | 1.219 | −0.396 | −1.216 | Venezuela | 0.689 | −1.275** | 1.092 | −1.167 | −2.083 |
| Netherlands | −1.028 | 0.525 | 0.381 | 1.377 | −1.244 | Thailand | 0.742 | 0.908 | 0.773 | 1.174 | −0.119 |
| South Korea | 0.602 | 0.793 | 0.544 | 1.458 | −0.381 | Philippines | 1.103 | −0.932*** | 0.619 | −1.506 | −3.121 |
| Austria | −0.426 | 0.547 | 0.278 | 1.964 | −1.629 | Bulgaria | −0.509** | −1.217*** | 0.436 | −2.793 | −5.088 |

**Note:** The table shows the OLS results for UIP Eq. (4), $s_{t+k} - s_t = \alpha + \beta(f_t^k - s_t) + \varepsilon_{t+k}$. The fifth and sixth columns in both panels show the $t$-statistics for beta being significantly different from zero and one, respectively. SE shows standard errors.
*** Denotes statistical significance at 1% level, respectively.
** Denotes statistical significance at 5% level, respectively.
* Denotes statistical significance at 10% level, respectively.

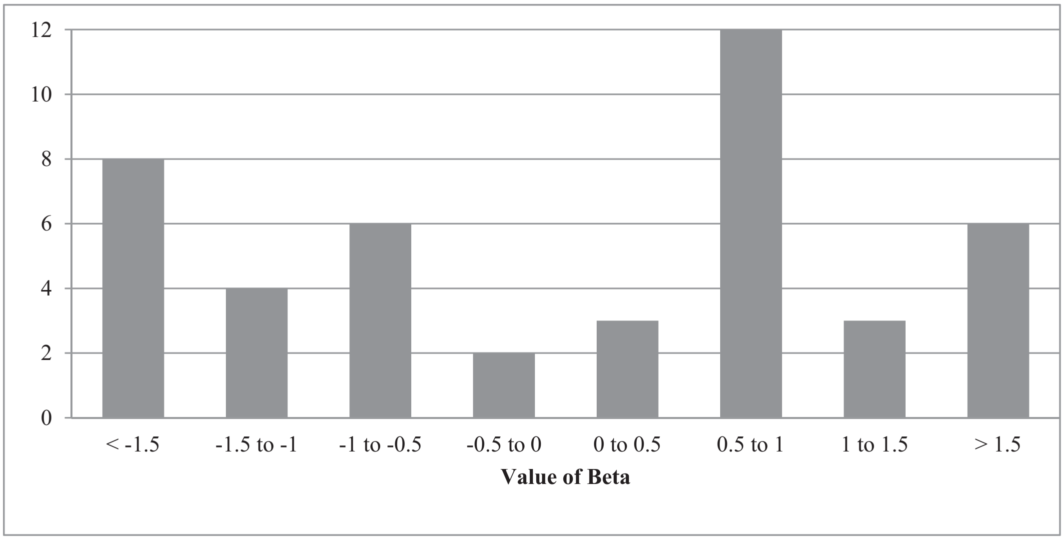


**Fig. 1.** Frequency distribution of beta coefficients in Table 2.

distribution (GED). The results for advanced and emerging countries are presented separately and Table 3.1 reports the CGARCH-M UIP results for advanced countries.

The intercept ($\alpha$) is significant at 5% level for Taiwan, Canada, Denmark and Singapore, implying the presence of a constant risk premium. One important observation is that the beta coefficient is noticeably negative and away from 1 only in case of two currencies now, that is, Canada and Norway as against Table 2 where it was significant for six currencies. The beta coefficients in Table 3.1 range from −0.618 to 1.431, with an average beta of 0.673, which is an improvement over an average beta of 0.544 in Table 2. As a consequence of including the risk premiums, the beta coefficients have moved closer to one for four out of these six currencies (i.e., Taiwan, Denmark, Hong Kong and Singapore). It suggests that in the presence of risk premium, there is a positive relation between exchange rate changes and the interest rate differentials, conforming to

**Table 3.1**
CGARCH-M Results for UIP for advanced countries.

| | $\alpha$ | $\beta$ | $t$, $\beta$=1 | $\theta$ | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ | $\gamma_4$ | $\gamma_5$ | $\gamma_6$ | $\gamma_2$=$\gamma_3$ = 0 | $\alpha$=$\theta$ = 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Australia | −0.252 | 0.920 | −0.134 | 0.630 | 0.006** | 0.698*** | 0.087 | 0.119 | −0.240** | −0.183** | 0.000 | 0.255 |
| UK | 0.284 | 0.889 | −0.214 | 0.388 | 0.002 | 0.595*** | 0.105* | 0.088 | −0.363** | 0.214 | 0.000 | 0.369 |
| Taiwan | −0.55** | 0.460 | −1.242 | −1.89*** | 0.005** | 0.265 | 0.068 | −0.097 | 0.284*** | 0.135 | 0.685 | 0.000 |
| Canada | −1.08*** | −0.62** | −2.064 | −2.29*** | 0.005*** | 0.863*** | 0.256*** | 0.225** | 0.122 | 0.348*** | 0.000 | 0.000 |
| Switzerland | −0.567 | 0.551 | −1.300 | −0.545 | 0.004** | 0.136 | 0.119 | 0.003 | 0.262** | −0.119 | 0.928 | 0.187 |
| France | 0.131 | 1.118 | 0.171 | 0.296 | 0.006*** | 0.540*** | 0.147 | 0.175*** | 0.126 | −0.223 | 0.001 | 0.435 |
| Czech Republic | 0.598 | 0.783 | −0.500 | −0.826 | 0.022*** | 0.769*** | −0.077* | 0.137** | 0.217* | 0.164 | 0.000 | 0.263 |
| Denmark | −0.689** | 1.267 | 0.276 | −1.51*** | 0.001 | 0.670** | 0.177** | 0.051 | −0.156** | 0.099 | 0.000 | 0.006 |
| Hong Kong | −0.470 | 0.864 | −0.106 | −0.88** | 0.007** | 0.828*** | 0.122** | 0.192** | 0.275** | −0.263** | 0.000 | 0.027 |
| Sweden | −0.490 | 0.745 | −0.401 | 1.107 | 0.008 | 0.728*** | 0.107 | 0.093 | 0.265** | 0.191 | 0.000 | 0.231 |
| Singapore | −1.43*** | 0.752 | −0.922 | −1.67*** | 0.010*** | 0.693*** | 0.089 | −0.086 | 0.117 | 0.317 | 0.000 | 0.000 |
| Germany | 0.403 | 0.826 | −0.298 | −0.228 | 0.009** | 0.511** | 0.097* | 0.112 | 0.193* | 0.165 | 0.000 | 0.259 |
| Japan | −0.313 | 0.657 | −0.427 | −0.467 | 0.003 | 0.360 | 0.119 | 0.046 | 0.261*** | −0.191 | 0.837 | 0.272 |
| Greece | 0.841 | 1.431 | 0.666 | 0.395 | 0.012*** | 0.823** | 0.172 | 0.118 | −0.165** | 0.491 | 0.000 | 0.223 |
| Norway | −0.388 | −0.39** | −2.405 | −2.62*** | 0.000 | 0.539*** | −0.024* | 0.056 | 0.215** | 0.283 | 0.000 | 0.011 |
| Italy | 0.685 | 0.862 | −0.217 | 0.718 | 0.002 | 0.911*** | 0.152** | 0.105 | 0.116 | 0.374*** | 0.000 | 0.358 |
| Spain | −0.311 | 0.273 | −0.645 | −0.397 | 0.001 | 0.863** | −0.19*** | 0.167** | −0.247** | 0.428 | 0.000 | 0.322 |
| Portugal | −0.547 | 0.926 | −0.144 | 0.675 | 0.005** | 0.753*** | 0.134 | 0.179** | 0.195* | 0.332 | 0.000 | 0.231 |
| Belgium | −0.112 | 0.429 | −0.430 | 0.113 | 0.001 | 0.605** | 0.001** | 0.015 | 0.158* | −0.198** | 0.000 | 0.347 |
| Netherlands | −0.428 | 0.687 | −0.869 | 0.622 | 0.010** | 0.680*** | −0.18*** | −0.137 | −0.226** | 0.287 | 0.000 | 0.311 |
| South Korea | −0.549 | 0.655 | −0.717 | −0.224 | 0.002 | 0.900*** | 0.155 | 0.205*** | −0.126 | 0.622** | 0.000 | 0.377 |
| Austria | −0.664 | 0.721 | −0.948 | −0.192 | 0.005** | 0.589*** | 0.099** | 0.125 | 0.178*** | −0.219** | 0.000 | 0.262 |

**Note:** The table shows the CGARCH-M results for UIP for advanced countries, as shown in Eq. (12). Column three shows the t-statistic for beta being significantly different from one.
*** Denotes statistical significance at 1% level, respectively.
** Denotes statistical significance at 5% level, respectively.
* Denotes statistical significance at 10% level, respectively.

**Table 3.2**
CGARCH-M Results for UIP for emerging countries.

| | $\alpha$ | $\beta$ | $t$, $\beta$=1 | $\theta$ | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ | $\gamma_4$ | $\gamma_5$ | $\gamma_6$ | $\gamma_2$=$\gamma_3$ = 0 | $\alpha$=$\theta$ = 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Russia | 0.392 | 0.926 | −0.092 | 0.357 | 0.009** | 0.888*** | 0.165** | 0.209** | 0.176*** | 0.276 | 0.000 | 0.475 |
| Colombia | 0.74*** | 1.228 | 0.107 | −2.33*** | 0.009*** | 0.847** | 0.107 | 0.161 | 0.013 | −0.342** | 0.000 | 0.000 |
| Turkey | −0.36** | 1.065 | 0.035 | −1.705** | 0.005 | 0.704** | 0.090* | 0.231** | 0.322*** | 0.228 | 0.000 | 0.000 |
| Brazil | 0.154 | 0.778 | −0.136 | −2.032** | 0.007*** | 0.976*** | 0.127 | 0.214*** | 0.243** | 0.443 | 0.000 | 0.002 |
| South Africa | 1.31*** | −1.29*** | −2.914 | −2.66*** | 0.012** | 0.912*** | 0.112 | −0.181 | 0.208*** | 0.298 | 0.000 | 0.000 |
| India | 0.424 | 1.583 | 0.699 | −0.99*** | 0.008** | 0.762*** | 0.243** | 0.222** | 0.272** | −0.182 | 0.000 | 0.005 |
| Argentina | 0.61** | 1.108 | 0.104 | −2.322** | 0.002*** | 0.852*** | −0.147** | 0.237** | 0.117** | 0.088 | 0.000 | 0.000 |
| Hungary | −0.81*** | −0.48*** | −3.524 | −1.17*** | 0.005** | 0.950** | 0.099 | 0.138 | 0.425*** | 0.411 | 0.000 | 0.000 |
| Chile | 0.929 | 0.978 | −0.016 | −1.906** | 0.001** | 0.824*** | 0.149*** | 0.127 | 0.516*** | 0.128 | 0.000 | 0.000 |
| Peru | −0.78*** | 0.268 | −0.478 | −1.564** | 0.001 | 0.724** | 0.137 | 0.191 | 0.328** | 0.297*** | 0.000 | 0.000 |
| Brunei Darussalam | 0.006 | 0.611 | −0.189 | −0.323* | 0.002** | 0.873** | 0.097** | 0.212* | 0.293** | −0.483** | 0.000 | 0.098 |
| Qatar | 0.413 | −0.73** | −2.259 | −0.46*** | 0.001*** | 0.927*** | 0.115** | 0.260** | 0.112 | −0.028 | 0.000 | 0.022 |
| Saudi Arabia | 0.63** | 0.619 | −0.299 | −0.50*** | 0.001*** | 0.748** | 0.095 | 0.193 | 0.096 | 0.115 | 0.000 | 0.000 |
| Indonesia | −0.179 | 0.487 | −0.681 | −1.077** | 0.002** | 0.899** | −0.114* | 0.208* | 0.265** | 0.198 | 0.000 | 0.015 |
| Mexico | −0.46** | 0.848 | −0.248 | −0.96*** | 0.003*** | 0.801*** | 0.121** | 0.164 | 0.372** | 0.371 | 0.000 | 0.000 |
| Poland | 0.652 | 1.178 | 0.542 | −0.317 | 0.001 | 0.926*** | 0.115 | 0.156 | 0.227*** | 0.223** | 0.000 | 0.623 |
| Malaysia | 0.842 | 0.847 | −0.237 | 0.193 | 0.007** | 0.815*** | 0.154** | −0.174 | 0.383*** | 0.542 | 0.000 | 0.549 |
| Morocco | −1.07*** | 1.171 | 0.123 | −1.258** | 0.002** | 0.830** | 0.145** | 0.235*** | −0.144** | 0.183 | 0.000 | 0.000 |
| Venezuela | 0.667 | 0.812 | −0.181 | −0.59*** | 0.002*** | 0.874*** | 0.127** | 0.175 | 0.087 | 0.297*** | 0.000 | 0.046 |
| Thailand | 0.898 | 0.937 | −0.115 | −0.165 | 0.006** | 0.987*** | −0.116** | 0.148 | 0.393** | 0.327 | 0.000 | 0.417 |
| Philippines | 1.02*** | 0.917 | −0.150 | −0.621* | 0.001 | 0.267 | 0.175*** | 0.152 | 0.066 | −0.116** | 0.000 | 0.071 |
| Bulgaria | −0.66*** | −0.44** | −2.385 | −1.68*** | 0.004*** | 0.917*** | 0.073 | 0.118 | −0.361** | 0.255 | 0.000 | 0.000 |

**Note:** The table shows the CGARCH-M results for UIP for emerging countries, as shown in Eq. (12). Column three shows the t-statistic for beta being significantly different from one.
*** Denotes statistical significance at 1% level, respectively.
** Denotes statistical significance at 5% level, respectively.
* Denotes statistical significance at 10% level, respectively.

the UIP theory. This is an additional improvement over the results for UIP when the risk premium component was not accounted for. Such a finding validates our argument that risk premium is the main factor responsible for UIP violation and including it in the main equation helps uncover the UIP puzzle.

To support the risk premium argument, the coefficient of time-varying risk premium ($\theta$) is significantly negative for those six advanced currencies for which UIP was violated. Further, the last column in Table 3.1 reports the Wald's test for no risk premium (both constant and time-varying) and shows that it is rejected for these six currencies. However, despite producing significant risk premium, results fail to conform to the UIP theory for Canada and Norway. The UIP failure for these countries could be explained based on Chen and Rogoff (2003), who argue that these are commodity currencies where changes in commodity prices also affect their exchange rates. They show a long-run run co-integrating relation between commodity prices and real exchange rates, and conclude that the standard exchange rate equations, if adjusted for commodity prices, could perform better for commodity currencies. However, doing so is beyond the scope of our study.

For the remaining advanced currencies, we do not find a significant risk premium which signals that UIP condition is not violated for these currencies. Such a finding is in line with the literature which highlights that developed and industrial countries are not characterized by risk premium (Domowitz and Hakkio, 1985; Baillie and Bollerslev, 1990). Nevertheless, there is an improvement while testing the UIP in the CGARCH-M framework compared to the OLS framework, a finding similar to Li et al. (2012), since the average beta coefficients are closer to one and conform more to UIP theory.

Table 3.2 shows the results for emerging countries. The intercept ($\alpha$) is significant at 5% level in case of half of the emerging countries signifying the constant risk premium. Compared to Table 2, the beta coefficient in Table 3.2 is negative and significantly different from one only for four currencies (South Africa, Hungary, Qatar, and Bulgaria). As against the UIP results in the OLS framework, the beta coefficients in the CGARCH-M model range from $-1.289$ to 1.583, with an average of 0.611. Of the 18 emerging currencies where the beta coefficients were significantly different from one leading to UIP failure in the OLS framework (shown in Table 2), the beta coefficients of 14 currencies have moved closer to one in the CGARCG-M model in the presence of risk premium. Therefore, in the presence of risk premium, the results are in line with the UIP, a phenomenon, more pronounced for emerging countries.

It is worth mentioning that the time-varying risk premium coefficient ($\theta$) is significantly negative for the 18 emerging currencies which witnessed the UIP violation. The last column shows the Wald's test to test the null of no risk premium which reveals that risk premiums are considerably different from zero for all 18 exchange rates at 10% significance level. These findings imply that risk premium play the main role in the UIP failure and including it in the main equation helps address the failure. However, we could not attribute the risk premium explanation to the failure of UIP for South Africa, Hungary, Qatar, and Bulgaria, which calls for finding other arguments apart from the risk premium theory.

Overall, the results in Tables 3.1 and 3.2 indicate that UIP hold for 20 out of 22 advanced currencies and for 18 out of 22 emerging countries in the CGARCH-M model, which is significant improvement over the results reported in Table 2 based upon the OLS model. Further, Fig. 2 shows the frequency distribution of the beta coefficients and reveals that almost 73% (32 out of 44) of the beta coefficients for both advanced and emerging countries lie between 0.5 and 1.5 with an average beta of 0.896. Since the beta coefficients move closer to one, it shows that high interest rate currency would depreciate against those with lower interest rates, which is in conformity with the UIP theory.[4] It shows that incorporating risk premium helps uncover the UIP failure.

Moreover, the negative $\theta$ coefficient is in line with the utility theory as in Li et al. (2012). It signifies that the depreciation of the home currency decreases with higher risk which leads to higher returns from holding such currencies. Other studies reporting the incidence of negative risk premiums are Tai (2003), Li et al. (2012), and Kumar and Trück (2014). Since the risk premiums for Russia, Poland, Malaysia and Thailand are not significant, the UIP condition is not violated for these currencies which strongly support our argument of risk premium being responsible for UIP failure.

A finding that risk premium is more significant for emerging countries is in line with Loring and Lucey (2013) and Li et al. (2012) who argue that risk premium holds more ground for emerging countries due to their higher risk characteristics relative to advanced countries. Aysun and Lee (2014) argue that a considerable part of UIP failure is explained by risk premium in case of both developed and developing countries. They further report highly significant risk premiums for more risky emerging currencies since emerging countries generally exhibit more conditional standard deviations relative to advanced countries, partly due to their lesser flexibility to carry effective fiscal and monetary policies, and less developed financial institutions (Hausmann et al., 2006; Li et al., 2012). Ichiue and Koyama (2011) argue that the results for UIP are more likely to hold in the presence of risk premium when there is high exchange rate volatility, especially in case of emerging countries. Therefore, we posit that risk premium must be considered both in theoretical as well as empirical models of the uncovered interest parity condition.

Our approach is similar to Li et al. (2012) who use CGARCH-M model for the first time to test the UIP theory. Though, they find significant risk premiums for many developed and developing currencies, however, the UIP failure persists even in the presence of risk premiums. We, on the other hand, address the UIP anomaly using risk premiums. An explanation for the improved results could be the use of more and newer data compared to Li et al. (2012) since the newer data includes a time span where capital has been more mobile. Under the increased capital mobility, the extent of interest rate arbitrage increases, therefore, the impact of factors other than risk premium on interest rate differentials would gradually weaken, providing results in conformity with the UIP (Berk and Knot, 2001). Therefore, our results provide support to the hypothesis of Berk and Knot (2001) who postulate that the introduction of new financial instruments and the elimination of capital

[4] Since almost all studies in the literature (including ours) rely on the approximate form of UIP, it might be possible that UIP failure arise from the model misspecification. Therefore, to avoid this problem, we re-estimate the UIP in its precise form, following Tang (2011) for the same sample of advanced and emerging countries. Specifically, we use fully modified OLS and dynamic OLS (details can be found in Pedroni, 2004). Though not reported, our results are qualitatively similar to those reported in Tables 2, 3.1, and 3.2 and we conclude that using the approximate form of UIP does not impact our results significantly.

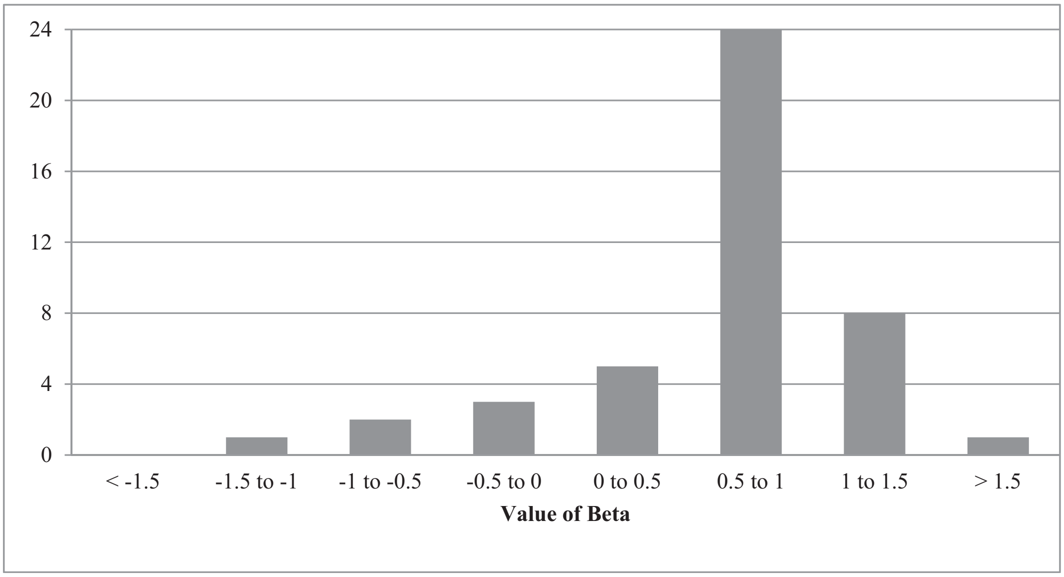


**Fig. 2.** Frequency distribution of beta coefficients in Tables 3.1 and 3.2.

controls have increased the capital mobility; consequently, the recent decades should provide stronger evidence in the favor of UIP than in earlier decades.

To support this claim, we undertake an additional analysis and show that the beta coefficients are found to be more positive and closer to one in the sample period especially after the GFC.[5] Using a Chow test, we find that all currencies exhibit a significant break around 2007–08, and based on Bussiere et al. (2018), we consider August 2007 as a common break date for all currencies. We show that after the crisis (post August 2007), the exchange rate changes are positively related to interest rate differentials and almost 89% of the beta coefficients lie between 0.5 and 1.5. This is our main contribution over and above Li et al. (2012) whose beta estimates were not unambiguously positive even after applying the GARCH-M model. We further conduct rolling regressions to show the evolution of beta coefficients and find that there is significant switch in the signs from negative to positive in the recent decade. In particular, the beta coefficients of all currencies (except Canada, Norway, South Africa and Qatar) stay positive after the financial crisis. Overall, we argue that these results are driven by time-varying risk premiums since it co-moves with interest rate differentials, more so in the post-crisis period.

Next, we turn to the discussion of the key parameters of the CGARCH-M model. In case of the long-run trend, the coefficient $\gamma_1$, representing the long-run average volatility, is positively significant for 13 advanced currencies and 18 emerging currencies, at 5% level, though the magnitude is near to zero. The permanent lagged volatility coefficient, $\gamma_2$, is very large and greatly significant for all advanced (except Taiwan, Switzerland, and Japan) and emerging currencies (except Philippines), which implies long-run persistence of volatility. On many occasions, the coefficient is closer to one, indicating higher trend persistence and slower convergence of permanent component of volatility to its mean. Our results are indicative of long memory in permanent volatility. Nearly 12 (14) of the 22 $\gamma_3$, the forecast error coefficients, are significant for advanced (emerging) currencies at 10% level of significance, representing the impact of progress of the permanent volatility component.

In short-run, the coefficient $\gamma_4$ represents the effect of a shock to the transitory constituent of our component GARCH framework which happens to be striking for seven advanced and nine emerging currencies. The parameter $\gamma_6$, implying the extent of memory in the short-run, is noteworthy for seven advanced and six emerging currencies. As explained above, the persistence of shock to the transitory component measured by $(\gamma_4 + \gamma_6)$ is lesser relative to the parameter $\gamma_2$, it indicates that in the long-run, the CGARCH framework is stable with a slow mean reversion. Hence, long-term volatility turns out to be relatively more persistent compared to short-term volatility. A higher component of the coefficient $\gamma_1$ implies that risk premium can be primarily attributed to shocks to macroeconomic forces, not to market sentiments. Such a finding is similar to that of Guimarães and Karacadag (2004), Pramor and Tamirisa (2006), and Li et al. (2012).

The coefficient of asymmetry $(\gamma_5)$ is significantly negative for six advanced currencies (Australia, UK, Denmark, Greece, Spain, and Netherlands) and two emerging currencies (Bulgaria and Morocco), which highlights the fact that volatility is driven more by unexpected home currency depreciation rather than unexpected appreciations. Similar results are reported by Li et al. (2012). On the contrary, 11 advanced and 15 emerging currencies exhibit significant positive $(\gamma_5)$ coefficient suggesting that an unexpected appreciation drives the volatility more than unexpected depreciation. The positive coefficients for

[5] These results have not been reported to conserve space. However, they are available upon request.

Switzerland, Japan, Russia, Turkey, South Africa, Hungary, and Mexico can be explained on the basis of carry trade literature since these are generally used as funding currencies and have a tendency to appreciate during high volatility periods. Our results support the notion that emerging currencies are more prone to carry trade and therefore exhibit more risk premium.

## 5. Conclusion and policy implications

We examine the role of risk premium in addressing the widely prevailing UIP failure for 44 advanced and emerging currencies for a data spanning over 47 years. Unlike advanced countries where beta coefficients are generally positive, the beta coefficients for emerging countries are negative which implies that a rise in interest rate differential will cause the expected exchange rate to decrease. On an average, 34% of the total beta coefficients lie between 0.5 and 1.5 for both advanced and emerging countries. It shows that UIP failure, or a negative relation between exchange rate changes and interest rate differential is a more common phenomenon in emerging countries relative to advanced countries.

One of the possible explanations for our results could be related to the risk premium theory. To account for a potential solution to the UIP failure, we use a CGARCH-M framework and show that the risk premium is significantly negative for many currencies considered, especially for emerging currencies. It means that investors are rewarded with both higher interest rate gains and currency appreciation gains for holding risky currencies. Moreover, except a few currencies, the beta coefficients move closer to one in the presence of risk premium. In fact, after incorporating the risk premium in the main equation, we observe that almost 73% of the beta coefficients range between 0.5 and 1.5 for both advanced and emerging countries, suggesting a positive relation between exchange rate change and interest rate differential, in lines with the UIP theory. It implies that risk premium is the main factor responsible for UIP failure and including it in the main equation helps uncover the failure. This is our key contribution to the literature.

Overall, our results suggest that considering risk premium in the main UIP equation provide encouraging results in that the UIP puzzle disappears especially for the emerging countries. Further, the results indicate that risk should be evaluated in terms of permanent (long-term) and transitory (short-term) components. Our estimates of the CGARCH-M model show that the permanent component has the most significant impact implying that shocks from long-term macroeconomic fundamentals (not the short-term market sentiments) are the primary determinants of exchange rate movements. An explanation for our results could be the use of more and newer data which includes a time span where capital has been more mobile as a result of the development of international financial markets.

Our results provide some important practical implications for international investors in the currency markets. One, since we observe a failure of UIP in many emerging and few advanced currencies, the monetary policy is believed to play a significant role in the foreign exchange markets. An expansionary (contractionary) monetary policy lead to decrease (increase) in the interest rate differential and may hamper (stimulate) inflow of foreign investments, consequently impacting the value of domestic currency. This is likely since many countries in our sample have an independent monetary policy and place less stringent controls on capital inflows. Two, there is a more tendency for international investors to engage in carry trade. The carry trade strategy intends to exploit the failure of UIP since the excess return to the carry trade increases as the high interest rate currency appreciates further against the low interest rate currency. However, as we show, in the post global financial crisis period, the incidents of UIP failure have reduced since the beta coefficients suggest a positive relation between exchange rate changes and interest rate differential. It may possibly be due to increased capital mobility under which the extent of interest rate arbitrage increases and the impact of factors other than risk premium on interest rate differentials would gradually weaken, providing results in conformity with the UIP. Therefore, it may be difficult to identify and reap the carry trade benefits in the recent times.

## Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intfin.2019.101135.

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